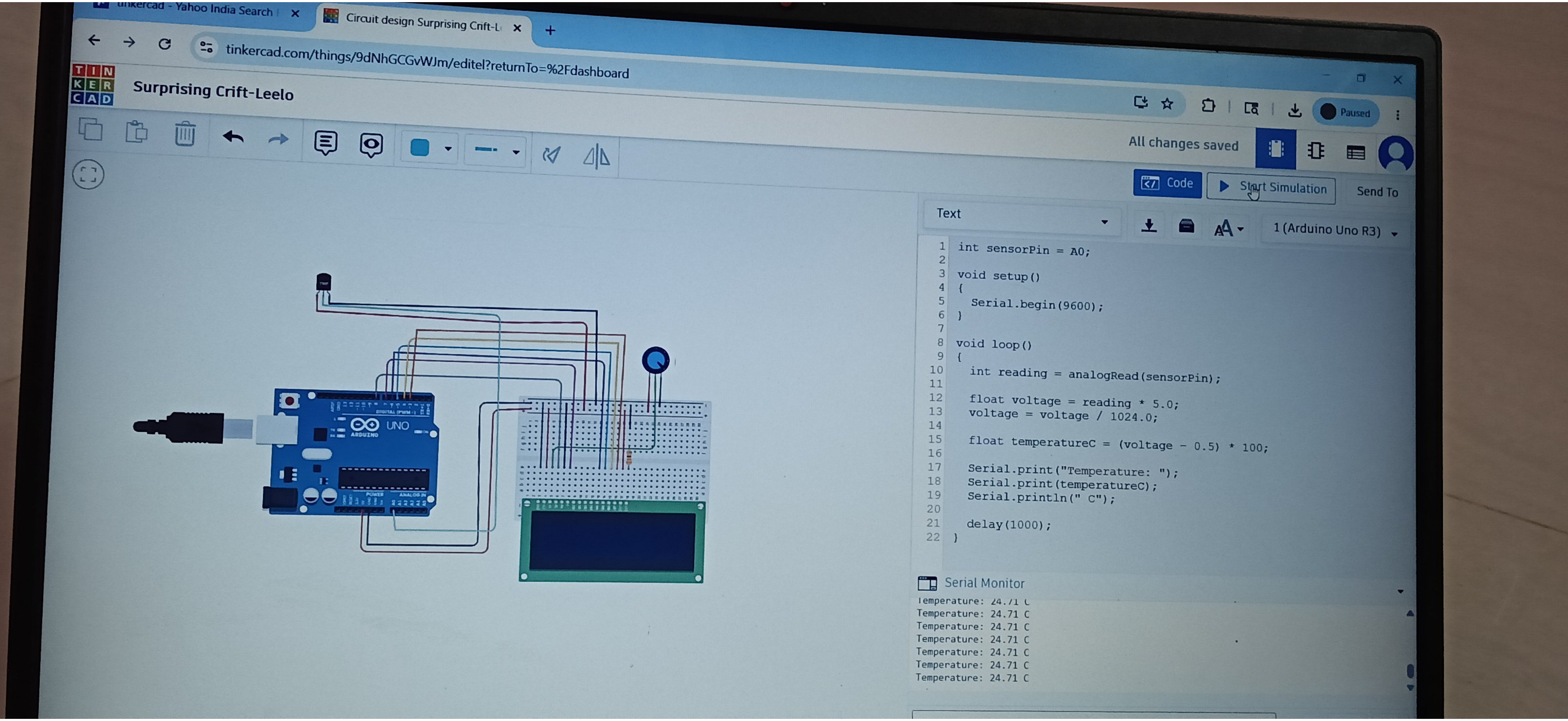
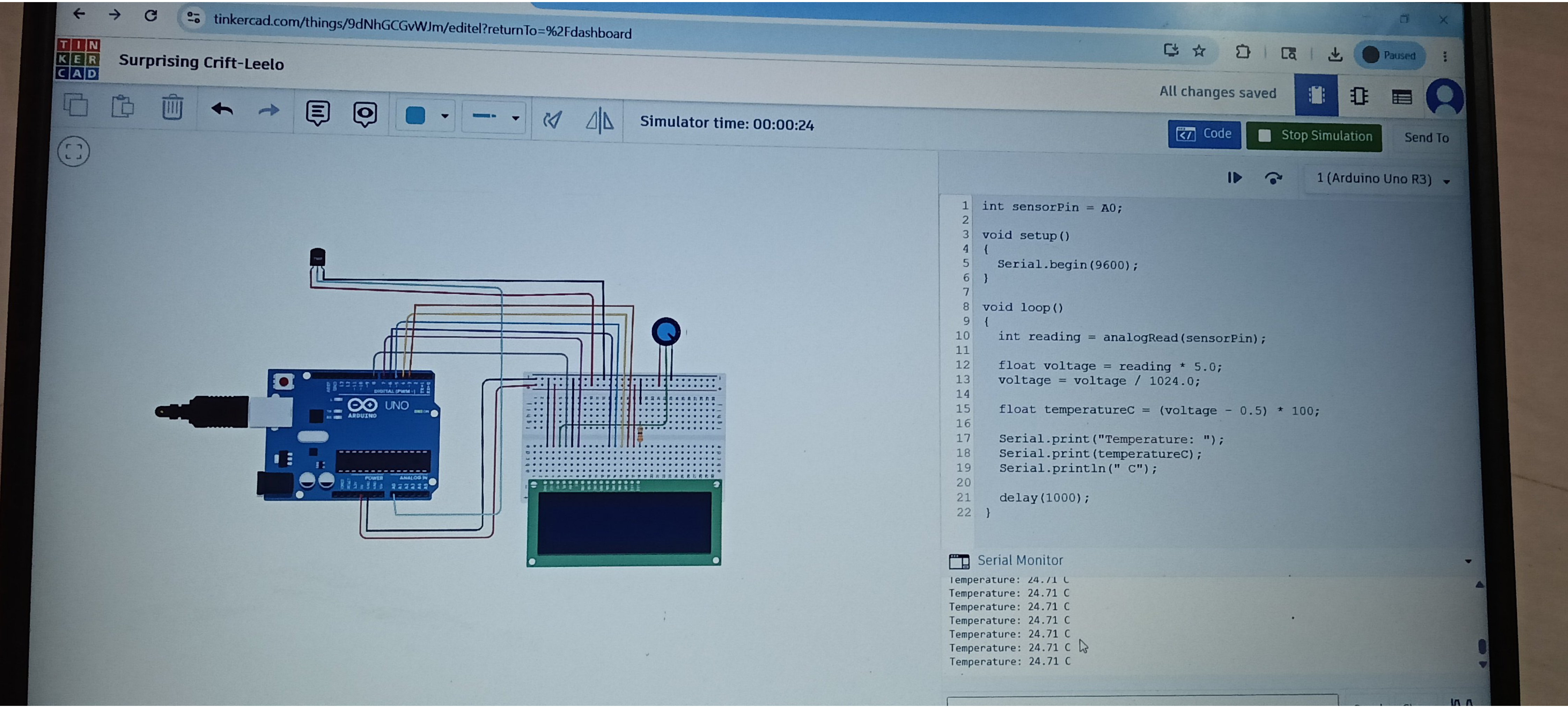
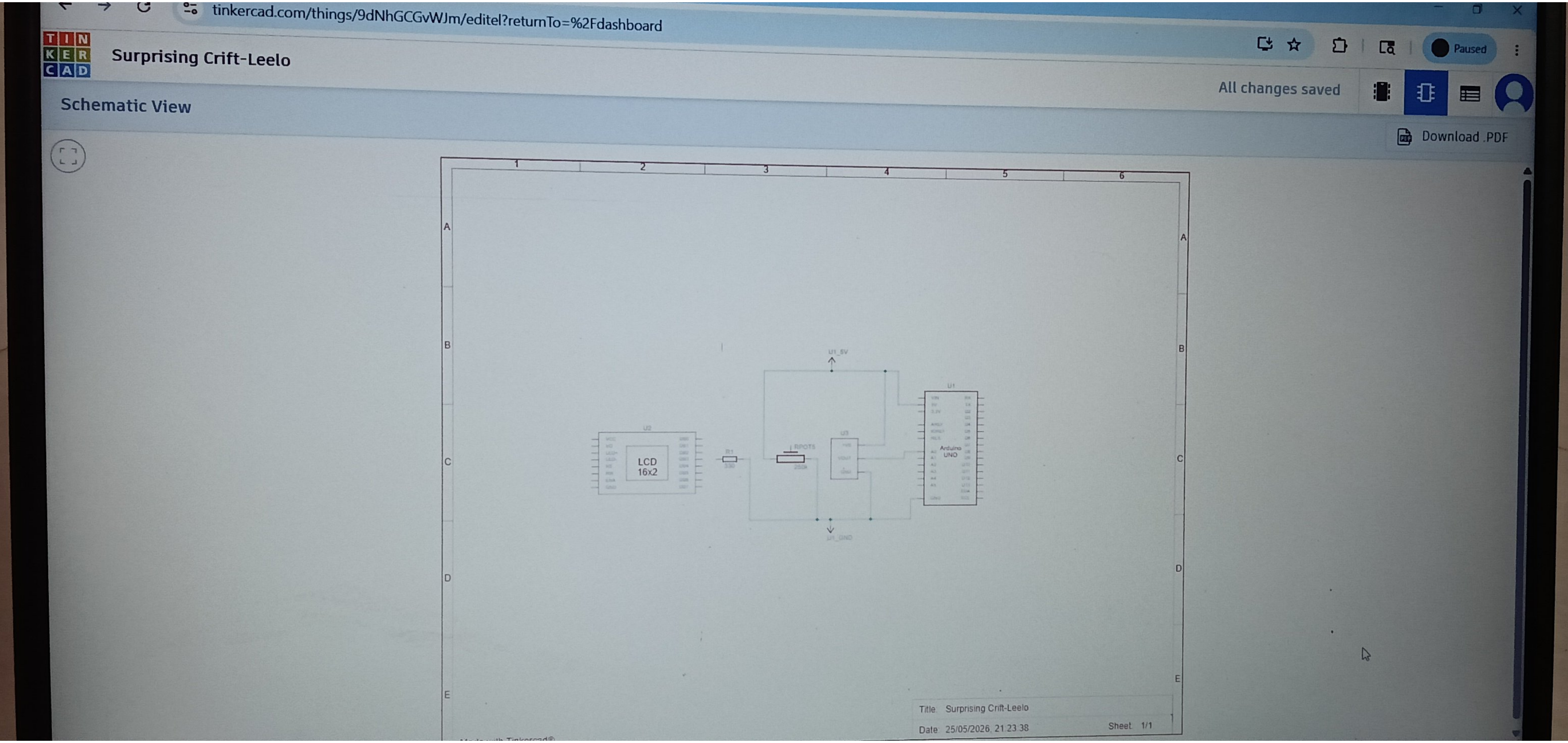

tinkercad - Yahoo India Search
Circuit design Surprising Crift-L
tinkercad.com/things/9dNhGCGvWJm/editel?returnTo=%2Fdashboard
TIN KER CAD
Surprising Crift-Leelo
Paused
All changes saved
Code
Start Simulation
Send To
Text
1 (Arduino Uno R3)
UNO
ARDUINO
int sensorPin = A0;
void setup()
{
Serial.begin(9600);
}
void loop()
{
int reading = analogRead(sensorPin);
float voltage = reading * 5.0;
voltage = voltage / 1024.0;
float temperatureC = (voltage - 0.5) * 100;
Serial.print("Temperature: ");
Serial.print(temperatureC);
Serial.println(" C");
delay(1000);
}
Serial Monitor
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C


tinkercad.com/things/9dNhGCGvWJm/editel?returnTo=%2Fdashboard
Surprising Crift-Leelo
All changes saved
Simulator time: 00:00:24
Code
Stop Simulation
Send To
1 (Arduino Uno R3)
int sensorPin = A0;
void setup()
{
Serial.begin(9600);
}
void loop()
{
int reading = analogRead(sensorPin);
float voltage = reading * 5.0;
voltage = voltage / 1024.0;
float temperatureC = (voltage - 0.5) * 100;
Serial.print("Temperature: ");
Serial.print(temperatureC);
Serial.println(" C");
delay(1000);
}
Serial Monitor
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
Temperature: 24.71 C
UNO
ARDUINO
Paused



tinkercad.com/things/9dNhGCGvWJm/editel?returnTo=%2Fdashboard
TIN KER CAD
Surprising Crift-Leelo
Paused
All changes saved
Schematic View
Download .PDF
LCD 16x2
Arduino UNO
Title: Surprising Crift-Leelo
Date: 25/05/2026, 21:23:38
Sheet: 1/1

tinkercad - Yahoo India Search | Circuit design Surprising Crift-L

tinkercad.com/things/9dNhGCGvWJm/editel?returnTo=%2Fdashboard

Paused

Surprising Crift-Leelo

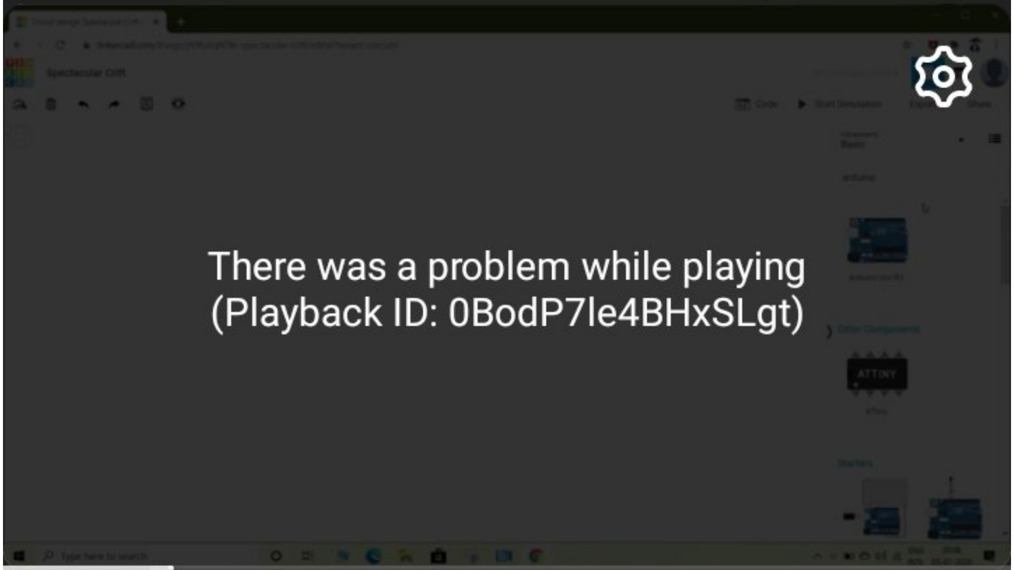
All changes saved

Component List

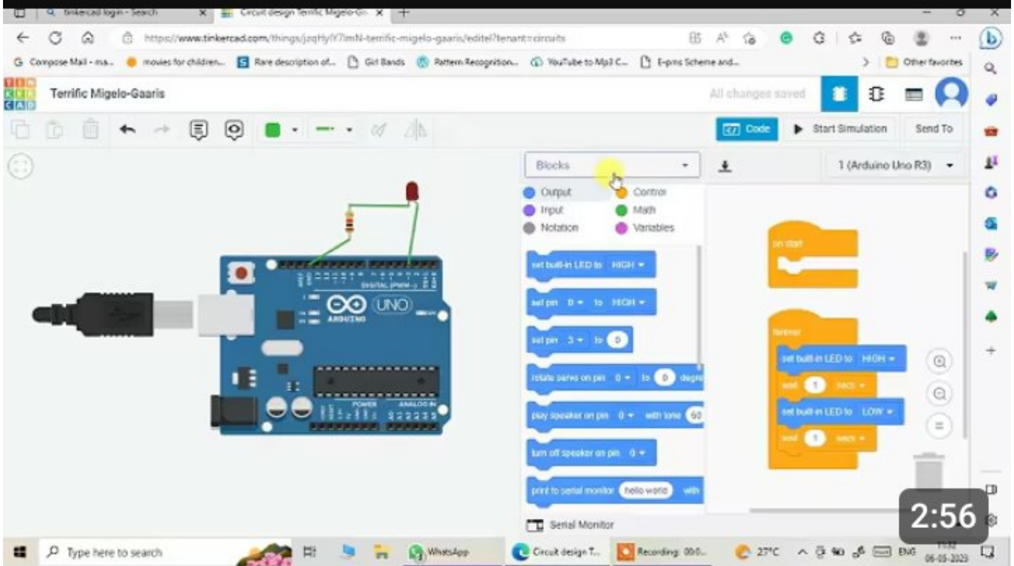
Download CSV

Name	Quantity	Component
U1	1	Arduino Uno R3
U2	1	LCD 16 x 2
Rpot5	1	250 kΩ Potentiometer
R1	1	330 Ω Resistor
U3	1	Temperature Sensor [TMP36]

9:44



All | From your search | From InnoventX | Ar



Led Blinking using Tinkercad

Manish_tutorials · 784 views · 3 years ago

TASK 3: IOT Prototype with Arduino

Project: Temperature and Humidity Monitoring System Using Arduino and DHT11 Sensor.

Introduction:

The Internet of Things (IOT) is a modern technology that allows devices and sensors to communicate with each other through the internet. IOT systems are widely used in homes, industries, healthcare, agriculture, and environmental monitoring. One of the most common and useful IOT applications is temperature and humidity monitoring.

Temperature and humidity are important environmental parameters that affect human, industrial processes, agriculture, storage systems, and weather conditions. Continuous monitoring of these parameters helps maintain proper environmental conditions and prevents damage caused by extreme temperature or humidity.

This project uses an Arduino UNO and a DHT11 sensor to measure temperature and humidity from the environment. The collected data is displayed on the serial monitor and can also be connected to cloud platforms for remote monitoring.

Objective:

The main objective of this Project is:

* To measure temperature and humidity using the DHT11 sensor.
* To display real-time environmental data using Arduino.
* To understand the working of IOT-based monitoring systems.
* To develop a low cost and efficient environmental monitoring system.

Components Required:

Name	Quantity	Components
U_1	1	Arduino UnoR3
U_2	1	LCD 16 X 2
Rpot 5	1	250 kΩ Potentiometer
R_1	1	330 Ω Resistor
U_3	1	Temperature Sensor [TMP36]

About DHT11 Sensor:

The DHT11 is a digital sensor used for measuring temperature and humidity. It is widely used in IOT projects because it is affordable, reliable, and easy to interface with Arduino.

Feautures of DHT11:

* Measures temperature from 0°C to 50°C
* Measures humidity from 20% to 90%.
* Low power consumption
* Easy digital output
* Compact and low-cost sensor.

The sensor has three pins:

1.) VCC → Power supply
2.) DATA → Sends DATA to ARDUINO
3.) GND → Ground connection.

Working Principle:

The DHT11 sensor senses the surrounding temperature and humidity and converts the data into digital signals. These signals are sent to the Arduino through the data pin.

The Arduino reads the sensor data using the DHT library and processes the value. The results are displayed on the serial monitor every few seconds.

If this system is connected to the internet using Wi-Fi modules like ESP8266 or NodeMCU the data can be uploaded to cloud platforms for remote monitoring.

Circuit Connections:

DHT11 to Arduino Connections

DHT11 Pin	Arduino Pin
VCC	5V
GND	GND
DATA	Digital Pin 2

Program Code:

```
int sensorPin = A0;
void setup()
{
  Serial.begin(9600);
}
void loop()
{
  int reading = analogRead(sensorPin);
  float voltage = reading * 5.0;
  voltage = voltage / 1024.0;
  float temperature = (voltage - 0.5) * 100;
  Serial.print("Temperature:");
  Serial.print(temperature C);
  delay(1000);
}
```

Explanation of the Code:

1.) Declaring the Sensor Pin:

```
int sensorPin = A0;
```

* This line declares a variable named sensorPin
* The temperature sensor output pin is connected to analog pin A0 of Arduino.
* Arduino reads analog signals from this pin

2.) Setup Function:

```
void setup()
{
  Serial.begin(9600);
}
```

Purpose:

⇒ The Setup() function runs only once when the Arduino starts.

Serial.begin(9600);

* Starts serial communication between Arduino and computer.

* 9600 is the baud rate [communication speed].

* This allows the temperature values to be displayed in the Serial Monitor.

3.) Loop Function:

⇒ void loop()

* The loop() function runs continuously

* Arduino repeatedly reads temperature data and displays it.

4.) Reading Sensor Value:

⇒ int reading = analogRead(sensorPin);

* analogRead() reads the analog voltage from pin A0.

* The value ranges from

0 → 0V

1023 → 5V

* Sensor reading stored in variable reading

5.) Converting Reading into Voltage:

```
float voltage = reading * 5.0;
voltage = voltage / 1024.0;
```

* Arduino converts analog values into voltage

Formula

$$V = \frac{\text{Reading} \times 5.0}{1024}$$

Working:

* reading * 5.0 converts sensor value into voltage.
* Dividing by 1024 gives the actual voltage value.

6.) Calculating Temperature:

```
float temperatureC = (voltage - 0.5) * 100;
```

* This formula converts voltage into temperature in Celsius.

Formula

$$T(^\circ C) = (V - 0.5) \times 100$$

Working:

* 0.5V is the offset voltage of the temperature sensor.
* Multiplying by 100 converts voltage into Celsius temperature.

Eg:

* If voltage = 0.75V

Then;

$$(0.75 - 0.5) \times 100 = 25^\circ C$$

7.) Printing Temperature:

```
Serial.print("Temperature:");
Serial.print(temperatureC);
Serial.println("C");
```

*These lines display the temperature value on the Serial Monitor.

8.) Delay Function:

```
delay(1000);
```

*Stops the program for 1000 milliseconds

*This means the temperature updates every 1 second.

Applications:

1.) Room temperature monitoring
2.) Weather monitoring systems
3.) Smart home automation
4.) Industrial temperature control
5.) IOT environmental monitoring systems.